

MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 35 | Date: 12-03-2026

Total Questions: 10 | Answer all questions.

Section 1: MCQ (5 Questions - 10 Marks)

Q1. [2 marks] (Easy)

Which data structure is characterized by LIFO (Last-In, First-Out) principle?

Options:

A) Queue B) Stack C) Linked List D) Tree

Q3. [2 marks] (Easy)

In a singly linked list, which pointer is typically NULL?

Options:

A) The head pointer B) The last node's next pointer C) The first node's data pointer D) The previous node's pointer

Q5. [2 marks] (Medium)

What will be the output of the following C code snippet? `int x = 10; int ptr = &x; ptr = 20; printf("%d", x);`

Options:

A) 10 B) 20 C) Address of x D) Error

Q7. [2 marks] (Easy)

Which operation adds an element to the rear of a queue?

Options:

A) Pop B) Enqueue C) Push D) Dequeue

Q9. [2 marks] (Medium)

What is the worst-case time complexity for searching an element in a sorted singly linked list?

Options:

A) $O(1)$ B) $O(\log n)$ C) $O(n)$ D) $O(n^2)$

Section 2: Short Answer (5 Questions - 25 Marks)

Q2. [5 marks] (Medium)

Explain the primary difference between a "primitive" and a "non-primitive" data structure with an example for each.

Q4. [5 marks] (Hard)

Compare and contrast arrays and singly linked lists based on memory allocation, insertion/deletion efficiency, and

random access capability.

Q6. [5 marks] (Medium)

Explain the concept of an Abstract Data Type (ADT). How does it differ from a concrete data structure?

Q8. [5 marks] (Hard)

Describe two common applications where a Stack data structure is primarily used.

Q10. [5 marks] (Hard)

You are given a singly linked list. Write down the conceptual steps (without actual code) to delete the node at the beginning of the list. Assume the list is not empty.

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